UNIVERSITY OF LEICESTER

CO2124

Muhammad Iftikhar

Lecture 8:
Mobile App Development (2)

UNIVERSITY OF LEICESTER

Announcement about the Class Test

- 25th & 26th March, 2026
- The test will be in-person closed book
- 1 hour fixed duration once the test starts. Once started the test must be completed
- This will be 30% on the module marks
- Venue: **Percy Gee Lab**
- There is no mock test since we have 4 quizzes already. The class test will follow the same format at the biweekly quizzes

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Announcement about the Class Test

Class Test The class test accounts for 30% of your overall semester mark. The test contains 45 questions, and the standard duration is one hour. Please read the essential information below carefully:

- **Students with Arrangements:** If you have a support plan or specific exam arrangements in place (e.g., extra time), we will contact you individually next week regarding your specific timing and location.

- **Group/Time Allocation:** Please attend according to your weekly lab session:
  - **Wednesday: 09:00 – 11:00 Labs:** Test runs 09:10 – 10:10.
  - **Wednesday: 11:00 – 13:00 Labs:** Test runs 11:10 – 12:10.
  - **Thursday: 09:00 – 11:00 Labs:** Test runs 09:10 – 10:10.
- *Please be seated in the lab at least 10 minutes before your start time. To avoid disruption, late arrivals will not be admitted.*

- **Location:** This test must be taken in the **Percy Gee labs only** during your allocated time. Attendance will be strictly monitored; submitting the test from outside the designated lab will be recorded as an absence.

- **Format:** The test will include MCQs, multiple-answer questions, fill-in-the-blank, and matching questions.

- **Academic Integrity:** This is a closed-book official university test. The use of mobile phones, textbooks, or the internet is strictly prohibited. Any violation of these rules will be regarded as cheating in accordance with university regulations.

- **Content Scope:** The test covers all material delivered this semester (CO2124), including all lab worksheets.

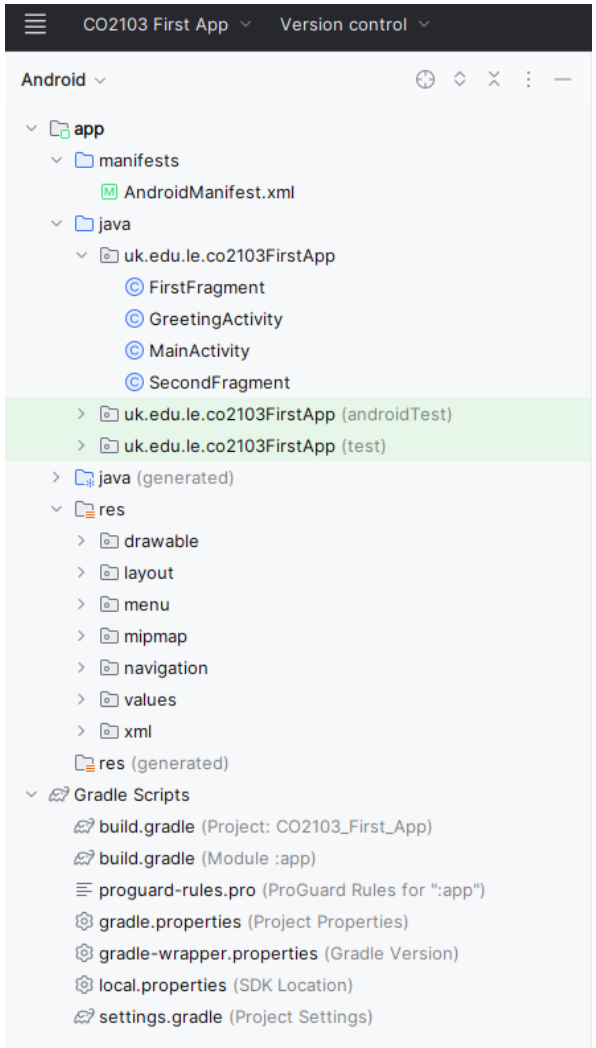
Previous Lecture,

- Android Architecture
- Android Components
- Activities Lifecycle

Today’s lecture includes,

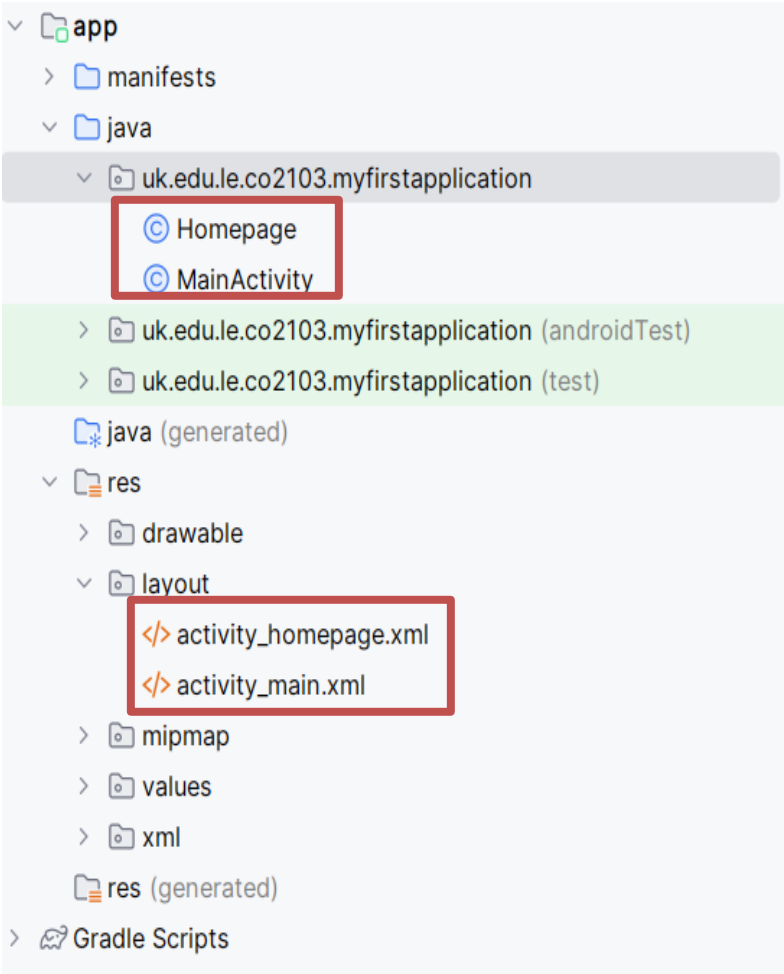
- App Structure
- Activities
- Layouts
- Intents
- Views
- Android Toast

App Structure



- **manifests**
  - Android Manifest file: vital for declaring your app's components, permissions, and other core configurations.
- **Java**
  - Java source code and test code
- **Res**
  - XML resources: layout, strings, drawable, dimensions, colours, etc.
- **Gradle Scripts**
  - Project
  - Module
  - Settings

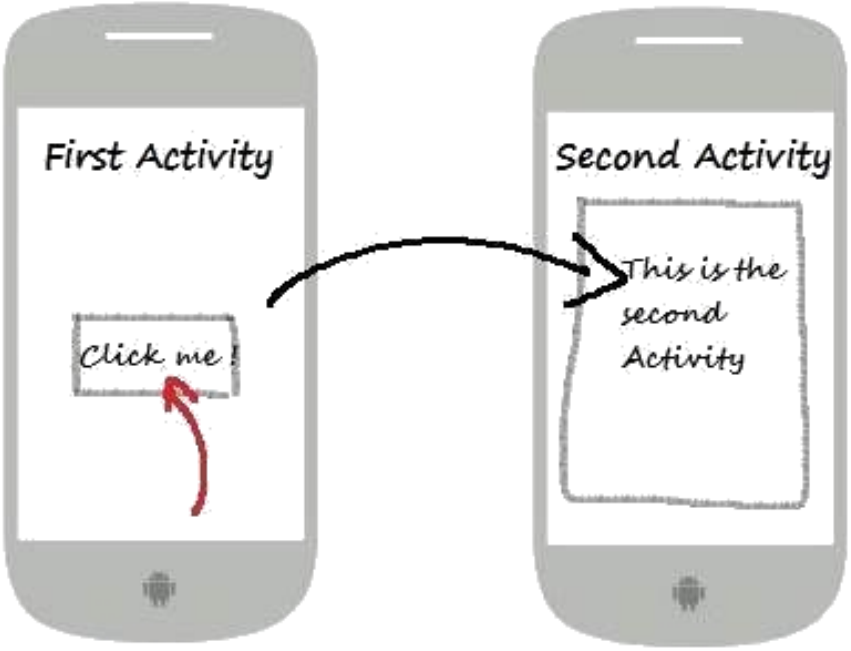
Activity XML Layout Files



- An activity may have an associated XML layout file. (see Figure)
- An activity can also create the layout programmatically (typically from inside the onCreate() method of the Activity subclass).
- However, in most cases it is easier to use a layout XML file. The layout XML file enables you to change the layout without having to change the code inside your Activity subclass.

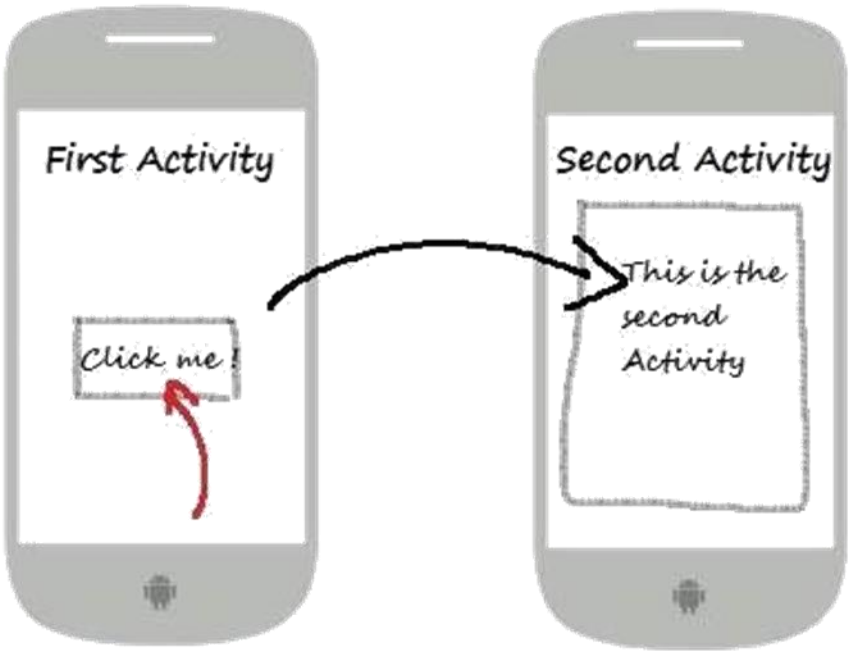
Activities

- An Activity is an application component
  - Represent tasks or actions in the real world, e.g., an activity for a list of recipes, getting the best route between two postcodes, booking a flight, etc.
- Typically fills the screen but can also be embedded in other Activity or appear as a floating window



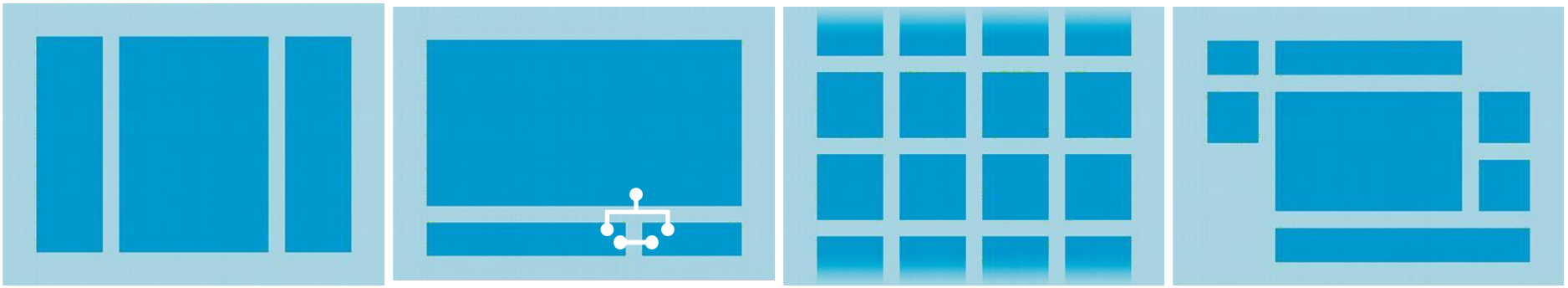
Activities

- Typically, one Java class file per Activity
  - With UI layout defined in one or more XML files (incl. variations)
- One or more activities loosely tied together make up an app
- Declared in Android Manifest file
  - There must be a **"main activity"**
  - Parent-child relationships between activities



Layouts

- ConstraintLayout: Positions UI elements using constraint connections to other elements and to the layout edges
  - Layouts are specific types of ViewGroups (subclasses of ViewGroup)
  - Can be in a row, column, grid, table, absolute



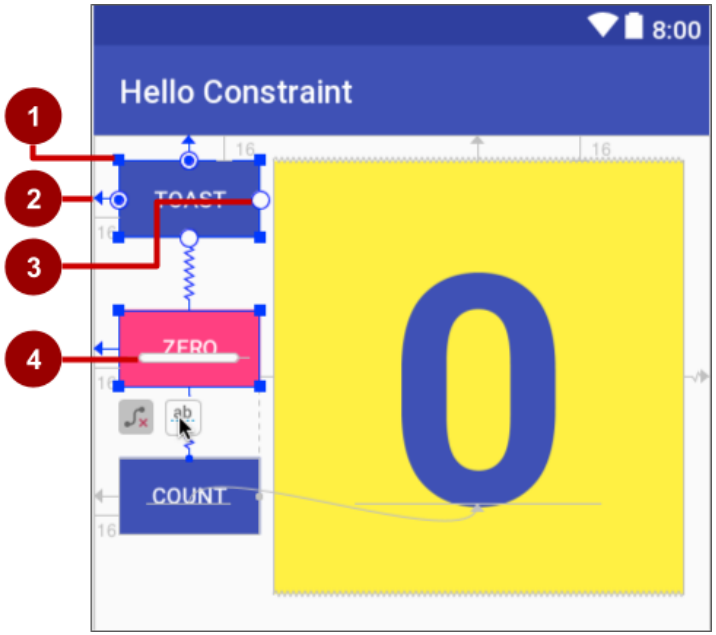
`LinearLayout` `ConstraintLayout` `GridLayout` `TableLayout`

- ScrollView: Contains one element and enables scrolling
- RecyclerView: Contains a list of elements and enables scrolling by adding and removing elements dynamically

[Images: Google Developers Training. Android Developer Fundamentals V2]

ConstraintLayout

- Default layout for new Android Projects
- Flexibility for layout design
- Provides constraints to position and align UI elements
- Constraints with respect to siblings / parent layouts



[Image: Google Developers Training. Android Developer Fundamentals V2]

Intents

- Description of an operation to be performed.
- An Intent object is used to request an action from another app component (e.g., another Activity) via the Android system
  - Start an activity upon a tap on a button, e.g., add a new item, book a flight, share a post, etc
  - Start a Service, e.g., start downloading a file in the background
  - Deliver Broadcast, e.g., Inform all apps that the battery is running low

– Explicit and Implicit intents

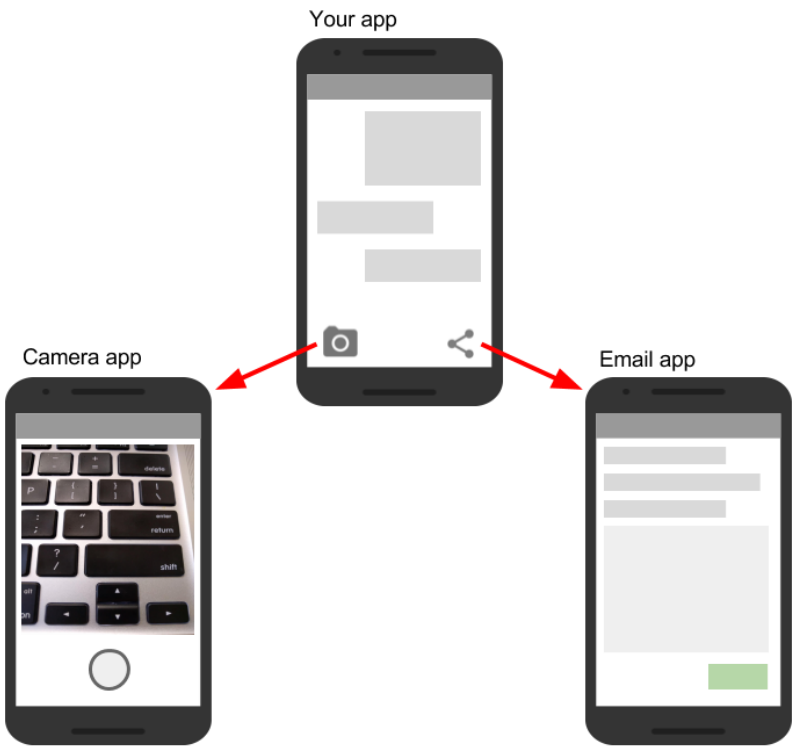
Intents

- Explicit
  - Explicit: Start a specific activity
    - often used within your own application since you will know the class name of the Activity that you would like to open.

```
Intent intent = new intent(welcomeActivity.this, orderActivity.class);
startActivity(intent);
```

Intents

- Implicit:
  - Ask the system to find a suitable Activity, e.g., to start a call




```
Uri uri = Uri.parse("tel:07477555333");
Intent it = new Intent(Intent.ACTION_DIAL, uri);
startActivity(it);
```

Views

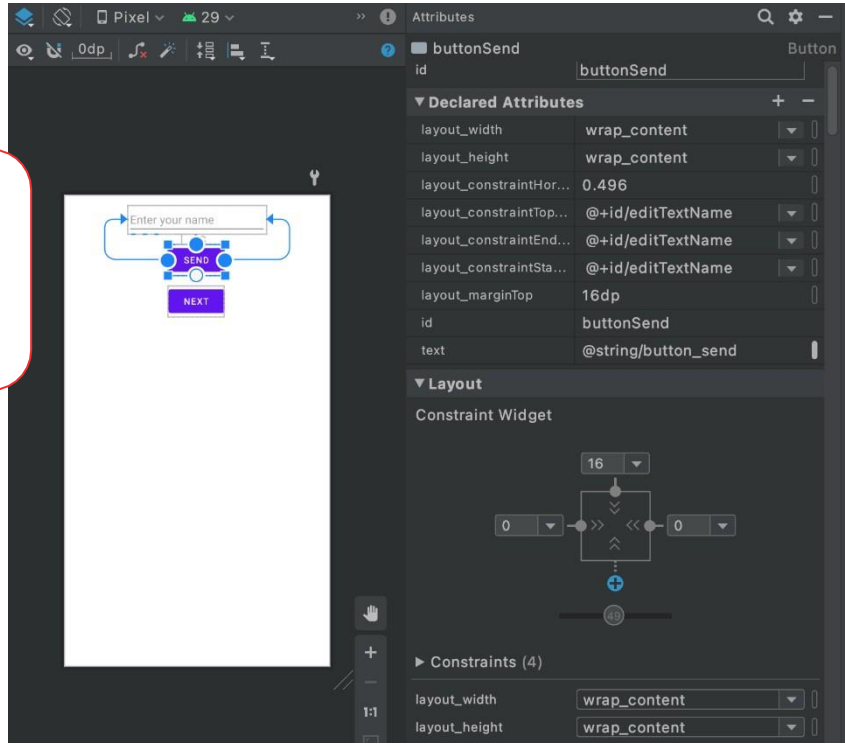
- Everything you see on the screen is a **View (**every UI element)
- Many different types of view (subclasses of **View**)
  - Display text (TextView class), edit text (EditText class)
  - Buttons (Button class)
  - Menus
    - Options menu: Global menu items for an app / activity
    - Context menu: Floating menu for long-click actions
    - Popup menu: Modal menu anchored to a specific **View**
  - Scrollable (ScrollView, RecyclerView)
  - Images (ImageView)
  - Other views: Checkbox, RadioButton, Slider, Switch
- Views can be grouped
  - Group views: ConstraintLayout and LinearLayout

View Attributes

- Color, dimensions, positioning (***Use device-independent measurements***)
- May have **focus**: ready to receive user input
- May be **interactive**: respond to user clicks
- May be **visible**: or not
- Relationships to other views
- Editable in XML or through Layout Editor

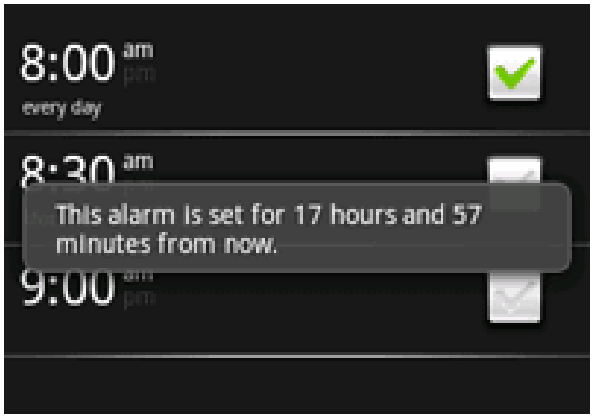
```
<Button
   android:id="@+id/buttonSend"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:layout_marginTop="16dp"
   android:text="@string/button_send"
   app:layout_constraintEnd_toEndOf="@+id/editTextName"
   app:layout_constraintHorizontal_bias="0.496"
   app:layout_constraintStart_toStartOf="@+id/editTextName"
   app:layout_constraintTop_toBottomOf="@+id/editTextName"
/>
```

Value, resource, or id reference



Android Toast

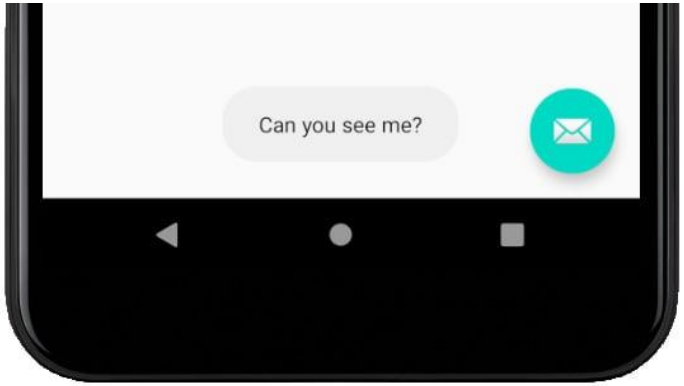
- An Android Toast is a small message displayed on the screen, similar to a tooltip or other similar popup notification.
- A Toast is displayed on top of the main content of an activity and only remains visible for a short time period.



Android Toast

- The method takes 3 parameters.
  - First the methods needs a Context object which is obtained by calling getApplicationContext().
  - The second parameter is the text to be displayed in the Toast.
  - The third parameter is the time duration the Toast is to be displayed.

```
Toast toast =
Toast.makeText(getApplicationContext(),
            "Can you see me?",
            Toast. LENGTH_SHORT);
toast.show()
```



Logging

- Logcat Panel shows information as your application runs.
- Use filters and search for tags to see what matters to you

```
FirstFragment
GreetingActivity
MainActivity
SecondFragment
uk.edu.le.co2103.firstapp (androidTest)
ExampleInstrumentedTest
uk.edu.le.co2103.firstapp (test)
java (generated)
res
drawable
layout
menu
mipmap
navigation
values
res (generated)
Gradle Scripts
build.gradle (Project: CO2103_First_App)
build.gradle (Module: CO2103_First_App.app)
```

```
40
41          Log.d(TAG, msg: "Adding Send button functionality");
42          Button button = (Button) view.findViewById(R.id.buttonSend);
43          button.setOnClickListener(new View.OnClickListener() {
44              @Override
45              public void onClick(View view) {
46                  Log.d(TAG, msg: "Executing buttonSend.onClick()");
47                  Intent intent = new Intent(getActivity(), GreetingActivity.class);
48                  EditText editText = (EditText) view.getRootView().findViewById(R.id.editTextName);
49                  String message = editText.getText().toString();
50                  if (message.toUpperCase().equals("WTF"))
51                      Log.wtf(TAG, msg: "What a Terrible Failure!");
52                  intent.putExtra(USER_NAME, message);
53                  startActivity(intent);
54              }
55          });
56      }
57  }
```

```
Logcat
Emulator Android_Accelerated_C | uk.edu.le.co2103.firstapp (7273) | Verbose | FirstFragment | Regex | Show only selected application
logcat
2021-03-14 16:11:24.857 7273-7273/uk.edu.le.co2103.firstapp D/FirstFragment: Executing buttonSend.onClick()
2021-03-14 16:11:24.857 7273-7273/uk.edu.le.co2103.firstapp E/FirstFragment: What a Terrible Failure!
2021-03-14 16:28:15.893 7273-7273/uk.edu.le.co2103.firstapp V/studio.deploy: Searching for class 'uk/edu/le/co2103/firstapp/FirstFragment' in the thread
2021-03-14 16:28:43.845 7273-7273/uk.edu.le.co2103.firstapp D/FirstFragment: Adding Send button functionality
2021-03-14 16:28:52.088 7273-7273/uk.edu.le.co2103.firstapp D/FirstFragment: Executing buttonSend.onClick()
2021-03-14 16:29:00.095 7273-7273/uk.edu.le.co2103.firstapp D/FirstFragment: Executing buttonSend.onClick()
2021-03-14 16:29:00.095 7273-7273/uk.edu.le.co2103.firstapp E/FirstFragment: What a Terrible Failure!
```

```
Show only selected application
Firebase
No Filters
Edit Filter Configuration
```

```
TODO | Terminal | Database Inspector | Profiler | 4: Run | Build | 6: Logcat | Event Log | Layout Inspector
Success: Operation succeeded (9 minutes ago)
44:22 LF UTF-8 4 spaces
```

See you in the next lecture…